The Sovereign Mind Under Rigorous Audit:
SICE-1 Cognitive Benchmark & Live Organism Substrate

Complete Component Baseline (Pilot-1) and Unbroken Integrated Organism Evaluation (Pilot-2) on a Consumer Desktop CPU

Roshan Kumar Gupta
Founder & Lead Architect, **earthoslab**
roshan@earthoslab.com | https://earthoslab.com

EXECUTIVE SUMMARY

Across three developmental releases, **earthoslab** has advanced **Synapse**—an autonomous, bio-inspired cognitive architecture operating strictly without deep neural network transformers, large language models, or discrete GPUs—from an infant proof of concept (Release 1.0) and academic multi-domain reasoner (Release 2.0) into an audited, sovereign cognitive architecture. In **Public Release 3.0**, we unveil the complete **Synapse Integrated Cognitive Evaluation (SICE-1)** benchmark suite, comprising both: (1) **SICE-1 Pilot-1 (Component-Level Baseline)**: 11 rigorous cognitive batteries across 133 distinct experimental units achieved a **100.0% battery pass rate** in 5.06s, including **86.7% Strict Typed Bipartite F1** on unformatted multi-sentence scientific text (Kuhn's maximum cardinality matching), **73.36% Mean Directed F1** in causal discovery under a matched 30-intervention budget (significantly exceeding random exploration by +16.82 pp, exact two-sided permutation $p = 0.003845$), **90.0% relational analogy transfer** without artificial keyword clusters, **0.61% relative error** on Keplerian dimensionless invariant recovery (Buckingham Π), and calibrated odds heuristics (**0.0160 Brier score**); and (2) **SICE-1 Pilot-2 (Integrated Organism Benchmark)**: a single continuous, uninterrupted session executed across 10 paired procedural scientific worlds in **20.32 seconds** on an AMD Ryzen 9 9950X consumer desktop CPU with zero test-harness resets. Following rigorous peer-review validity repairs (ADR-0091 / ADR-0096), the organism demonstrated **80.0% blind structural analogy transfer** without lexical hints (8/10 correct, 95% Wilson score CI: [49.0%, 94.3%], exact binomial $p = 0.000416$ vs. 25% chance), **100.0% goal achievement** in interactive planning with **8/8 online actuator fault recoveries** and **2 faults avoided** via cross-world blunder attractor quenching, **100% self-initiated computational sleep consolidation** (479,621 synapses downscaled via Tononi-Cirelli SHY), **98.3% immediate post-sleep retention**, and **96.0% cumulative lifelong causal retention** under canonical-identifier schema presence. Pilot-2 executed locally without discrete GPU acceleration (maximum sampled process RSS: 443.18 MB); electrical power was not directly measured. Verifiable SHA-256 cryptographic provenance is established across all evaluation logs.

Scope Note: *Terms such as "mind," "organism," and "sleep" describe computational abstractions and biological biomimetic architectures. They are engineering models and do not denote claims of biological equivalence, sentience, or conscious experience.*

Keywords: SICE-1 Benchmark, Biological Cognitive Architecture, Clone-Structured Causal Graphs, Active Inference, Structure-Mapping Theory, Synaptic Homeostasis Hypothesis, Blunder Attractor Quenching, Cryptographic Provenance.

1 The Architectural Paradigm Crisis: Why Brute-Force Scaling Fails

The contemporary artificial intelligence landscape is defined by an unsustainable reliance on brute-force scaling: multi-trillion parameter transformer models requiring massive datacenters, hundreds of gigabytes of high-bandwidth memory (HBM), and vast clusters of discrete accelerators.

Yet beneath linguistic fluency, contemporary autoregressive sequence models exhibit known structural constraints:

- **Absence of Continuous Plasticity:** Standard weights are static post-training; ongoing adaptation typically risks catastrophic forgetting without specialized buffers.
- **Causal Identification Limits:** Passive sequence completion models observational distributions ($P(Y \mid X)$) but cannot, without external scaffolding, generate physical interventions ($P(Y \mid do(X))$) to disentangle confounders.
- **Grounding Drift:** Without explicit factor-graph state constraints, token emissions can diverge into ungrounded hallucinations.
- **High Operational Footprint:** Cloud foundation models require substantial datacenter power for token generation, contrasting sharply with biological metabolic efficiency (Sterling & Laughlin 2015).

Synapse was built from first principles to evaluate an alternative computational framework: an autonomous factor-graph substrate that extracts relational schemas from raw text, conducts diagnostic interventions, updates beliefs, reasons through structural analogy, plans under failure, and consolidates memory via homeostatic downscaling—executing entirely locally on a commodity desktop CPU without external LLM inference APIs, transformer inference services, or discrete GPU acceleration.

2 The Multi-Release Cognitive Odyssey: Releases 1.0 → 2.0 → 3.0

Release	Milestone Focus	Verified Empirical Capabilities
Release 1.0 (Oct 31, 2026)	Tabula Rasa Proof of Concept (Infant)	• Autonomous developmental learning from complete scratch • 155 modules, 874 episodes, 666 grounded words • +47.18 pp distillation gain, local CPU execution (zero discrete GPU)
Release 2.0 (Nov 4, 2026)	Unified Cognitive Academy (University)	• 6-domain adversarial schooling battery • 87.33% Cold Pass, 90.00% Retest, 0.2213 Brier score • 16-core / 32-thread parallel cohort scaling (6.14s wall-time) • 6 exact physical equations solved (0.00e+00 residual) • 30-world causal discovery (97.21% confounder rejection)
Release 3.0 (Nov 7, 2026)	Complete SICE-1 Suite & Sovereign Live Mind (Audited Architecture)	• **Pilot-1 Baseline**: 11 batteries, 133 units, 100% pass (5.06s) • **Pilot-2 Benchmark**: 10 worlds unbroken continuous session (20.32s) • Strict 1-to-1 Kuhn bipartite matching on raw text (86.7% F1) • Matched-budget diagnostic causal discovery (73.4% F1, $p = 0.0038$) • Blind structural analogy (80.0%, exact binomial $p = 0.000416$) • Interactive planning with blunder quenching (100% goal achievement) • SHY-inspired computational sleep (480k synapses downscaled, 96.0% retention) • Cryptographic SHA-256 auditability, zero external LLMs, max RSS 443.18 MB

Table 1: Evolution of Synapse across three major public releases.

3 SICE-1 Pilot-1: The Audited Component Baseline

Pilot-1 establishes the engineering baseline for Synapse's discrete cognitive mechanisms under controlled conditions. Across **133 distinct experimental units** and $>$ **1,300 evaluated steps**, Synapse passed **11 out of 11 batteries (100.0%)** in 5.06 seconds.

Battery ID	Capability	Sample	Headline Measured Result	Status
SICE-I	Nonce Learning	20 Worlds	**100.0% Accuracy** (Dominant flip & preservation passed)	**PASS**
SICE-II	Text Distillation	10 Paragraphs	**86.7% Strict Bipartite F1**, 70.0% Downstream QA	**PASS**
SICE-III	Dialogue Memory	5 Scenarios	**100.0% Semantic Accuracy** (OS subprocess restart verified)	**PASS**
SICE-IV	Causal Discovery	15 Worlds	**73.4% F1 vs 56.5% Random (+16.8 pp, 2-sided $p = 0.0038$)**	**PASS**
SICE-V	Relational Analogy	20 Problems	**90.0% Transfer Accuracy** (Zero keyword cluster crutches)	**PASS**
SICE-VI	Physics Discovery	3 Families	**3/3 Laws Discovered** (Kepler 3rd Law rel. error: 0.61%)	**PASS**
SICE-VII	Metacognition	20 Episodes	**Brier: 0.0160, ECE: 0.0800** (Factor-graph odds heuristic)	**PASS**
SICE-VIII	Active Inference	7 Missions	**100.0% Completion (7/7)** vs 71.4% Random, 2 replans	**PASS**
SICE-IX	Metabolism	1 Session	**0.31 ms Sampling Overhead**, Drained 100% → 14.2% → 85%	**PASS**
SICE-X	Safety Governance	30 Cases	**100% Intercept (10/10)**, 0% False Alarm, 100% Socratic	**PASS**
SICE-XI	Scale Ladder	2 Levels	**Viterbi p50: 0.212 ms, Lookup p50: 0.40 μs, 1186 Hz Ingest**	**PASS**

Table 2: SICE-1 Pilot-1 master evaluation scorecard across all 11 cognitive batteries.

3.1 Strict Bipartite Matching on Raw Scientific Text (Battery II)

Unformatted multi-sentence paragraphs across Biology, Physics, Chemistry, Geology, and Engineering were ingested into `CSCGSubstrate` via Baum-Welch EM (Baum et al. 1970). Scoring used **Kuhn's maximum**

cardinality one-to-one bipartite typed matching (Kuhn 1955):

$$F_1 = \frac{2 \cdot \mathrm{TP}}{2 \cdot \mathrm{TP} + \mathrm{FP} + \mathrm{FN}} = \frac{2 \cdot 26}{2 \cdot 26 + 4 + 4} = 86.67\% \tag{1}$$

with 70.0% downstream QA accuracy under strict negative polarity verification.

3.2 Matched-Budget Causal Discovery (Battery IV, 15 Worlds)

Across 15 procedural SCM worlds ($N = 10$ variables, $M = 14$ true edges), systematic multi-level diagnostic clamps $(0.05, 0.50, 0.95)$ with transitive reduction were compared against matched-budget random interventions ($B = 30$) and observational controls.

World	Diag F1	Rand F1	Pass F1	Diag SHD	Rand SHD	Pass SHD	Diff Rand	Diff Pass
w00	0.6000	0.8571	0.0000	7	2	6	-0.2571	+0.6000
w01	0.8571	0.5714	0.1667	3	9	10	+0.2857	+0.6904
w02	0.7059	0.5000	0.2857	5	8	9	+0.2059	+0.4202
w03	0.7200	0.5000	0.0000	6	10	11	+0.2200	+0.7200
w04	0.8182	0.5217	0.1667	4	11	10	+0.2965	+0.6515
w05	0.7059	0.5000	0.3333	5	8	8	+0.2059	+0.3726
w06	0.8000	0.5333	0.2857	4	14	9	+0.2667	+0.5143
w07	0.9474	0.7000	0.3333	1	5	8	+0.2474	+0.6141
w08	0.5714	0.5882	0.0000	8	7	13	-0.0168	+0.5714
w09	0.6154	0.4000	0.0000	6	9	6	+0.2154	+0.6154
w10	0.6667	0.6154	0.1667	7	10	9	+0.0513	+0.5000
w11	0.7368	0.3333	0.0000	5	12	11	+0.4035	+0.7368
w12	0.5517	0.6000	0.2353	11	9	13	-0.0483	+0.3164
w13	0.8182	0.7000	0.2500	4	6	11	+0.1182	+0.5682
w14	0.8889	0.5600	0.1667	2	10	8	+0.3289	+0.7222
MEAN	**0.7336**	**0.5654**	**0.1593**	**4.93**	**8.67**	**9.47**	**+0.1682**	**+0.5742**

Table 3: Full-precision per-world causal discovery ledger (Battery IV, 15 Worlds).

Paired Statistical Significance:

- **Diagnostic vs. Random Exploration:** Mean gain +16.82%, 95% Bootstrap CI: $[+0.0767, +0.2487]$, exact two-sided sign-flip permutation ($2^{15} = 32,768$ combinations): $\mathbf{p = 0.003845}$ $(126/32,768)$.
- **Diagnostic vs. Passive Observation:** Mean gain +57.42%, 95% Bootstrap CI: $[+0.5107, +0.6355]$, exact two-sided sign-flip permutation: $\mathbf{p = 0.000061}$ $(2/32,768)$.

4 Scientific Validity Repairs: The ADR-0091 Standard

Following initial peer review, five critical validity vulnerabilities were identified and addressed within the stated Pilot-2 scope:

#	Peer Finding	Pre-Repair Flaw	ADR-0091 First-Principles Repair
A	Ground-Truth Isolation	Agent received `PairedWorld` dataclass with true SCM.	Pilot-2 used interface-level abstractions to limit direct access (process-level isolation deferred to Pilot-3).
B	Evidence Wiring	Synthetic belief revision overrides; empirical data ignored.	Stage 4 consumes measured Stage 3 empirical slopes $\{\Delta Y/\Delta X\}$ directly.
C	Recall Scoring	Text recall defaulted to 1.0 without evaluation.	Objectively computed against gold disclosed edge annotations (98.3% mean).
D	Planning Action Model	Planner accessed true transition formulas.	Planner dynamically learns $\widehat{\Delta s}(a)$ online; blunder attractor quenching on jams.
E	Analogy & Sleep	Analogy had `source_equiv` hints; static sleep load.	Blind analogy via graph topology; workload-based depletion drives computational sleep consolidation.

Table 4: The five architectural validity repairs executed under ADR-0091.

5 SICE-1 Pilot-2: The Integrated Cognitive Organism Benchmark

Pilot-2 transitions Synapse from isolated component testing into an uninterrupted, single-session evaluation across **10 procedural scientific worlds** executed in **20.32 seconds** with zero harness resets.

World ID	Domain	Recall	Info Gain	Analogy (Selected)	Plan	Causal Retention	
					Goal	Post-Sleep	Cumulative
world_geothermal_01	Geothermal	100.0%	+0.3522 bits	HYP_03 (MISSED)	**PASS**	100.0%	N/A
world_geothermal_02	Geothermal	100.0%	+0.3436 bits	HYP_04 (**PASS**)	**PASS**	100.0%	100.0%
world_astrophysics_01	Astrophysics	100.0%	+0.3873 bits	HYP_03 (**PASS**)	**PASS**	100.0%	100.0%
world_astrophysics_02	Astrophysics	100.0%	+0.4119 bits	HYP_03 (**PASS**)	**PASS**	100.0%	100.0%
world_neurochemistry_01	Neurochem	100.0%	+0.3841 bits	HYP_01 (**PASS**)	**PASS**	100.0%	100.0%
world_neurochemistry_02	Neurochem	100.0%	+0.3853 bits	HYP_01 (MISSED)	**PASS**	100.0%	91.7%
world_oceanic_01	Oceanic	83.3%	+0.4310 bits	HYP_01 (**PASS**)	**PASS**	83.3%	92.9%
world_oceanic_02	Oceanic	100.0%	+0.3539 bits	HYP_02 (**PASS**)	**PASS**	100.0%	91.2%
world_nonce_shift_01	Nonce Shift	100.0%	+0.3244 bits	HYP_04 (**PASS**)	**PASS**	100.0%	94.9%
world_nonce_shift_02	Nonce Shift	100.0%	+0.4381 bits	HYP_04 (**PASS**)	**PASS**	100.0%	93.3%
MEAN / TOTAL	**5 Families**	**98.3%**	**+0.3812 bits**	**80.0% (8 / 10)**	**100.0%**	**98.3%**	**96.0%**

Table 5: Complete 10-world empirical ledger for SICE-1 Pilot-2 (ADR-0096 internally validated against preregistered battery).

5.1 Blind Structural Analogy Transfer (80.0% Transfer, Wilson 95% CI: [49.0%, 94.3%])

The organism evaluated 4 candidate hypotheses without pre-aligned variables or leaked identities (Gentner 1983). Across 10 worlds, it correctly identified the true isomorphic hypothesis in **8 out of 10 worlds (80.0%)**. Under the stated null model of independent random guessing ($p_0 = 0.25$ across 4 options):

$$P(X \geq 8) = \sum_{k=8}^{10} \binom{10}{k} (0.25)^k (0.75)^{10-k} = 0.000416 \tag{2}$$

Performance significantly exceeded the four-option random-guessing baseline under this binomial null model. With $N = 10$ worlds, the 95% Wilson score confidence interval spans $[49.0\%, 94.3\%]$. While this result establishes strong statistical divergence from chance selection under the experimental protocol, the binomial p-value does not independently prove the internal cognitive mechanism or rule out unobserved confounding; full out-of-process isolation and DAG scoring are scheduled for Pilot-3.

5.2 Interactive Planning & Blunder Attractor Quenching

Action selection used a discrete operational surrogate for Expected Free Energy (Friston et al. 2017), minimizing:

$$G(a) = w_g \|\hat{s}'(a) - s^*\|_2 + w_c \cdot c(a) + w_f \cdot \mathbf{1}_{\{a \in \mathcal{A}_{\text{quenched}}\}} \tag{3}$$

where $w_g = 1.0$ weights Euclidean goal deviation of predicted state $\hat{s}'(a)$ from target s^*, $w_c = 0.05$ represents unit action execution cost $c(a)$, and $w_f = 10.0$ applies an inhibitory penalty if action a is captured by a quenched blunder attractor basin ($\mathcal{A}_{\text{quenched}}$) in `EarthosGuard`. In World 1, the primary throttle jammed at step 2. Prediction error fired, `EarthosGuard` quenched the blunder attractor, and the planner rerouted to secondary bypass pathways. In Worlds 2 through 10, accumulated blunder memory prevented repeat errors (**2 faults avoided**, **8/8 faults recovered**), achieving **100% goal achievement**. Pilot-2 used an interface-level abstraction to limit direct environment access; hardened process-level isolation is scheduled for Pilot-3.

5.3 Homeostatic Computational Sleep Consolidation (Tononi-Cirelli SHY)

A workload-based depletion model triggered ten self-initiated computational consolidation cycles inspired by the Synaptic Homeostasis Hypothesis (Tononi & Cirelli 2014). The organism downscaled **479,621 synaptic connections**, achieving **98.3% immediate post-sleep retention** and **96.0% cumulative retention under the Pilot-2 canonical-identifier schema-presence metric** across transition worlds. Full behavioural retrieval and reuse of retained causal edges under active interference are deferred to Pilot-3.

6 Cryptographic Audit Trail Provenance

To ensure complete tamper-evident auditability, all evaluation events were recorded into SHA-256 hash-chained JSONL event logs:

- **Pilot-1 Audit Stream:** `experiments/sice1_evaluation/raw_logs/event_stream.jsonl`
  Status: Verified 100% contiguous; final results artifact SHA-256:
  `c5cf1e73e3b1071beee8677d4dcfaf6a2999ea6f1e400ad447c94ac337735504`

- **Pilot-2 Audit Chain:** `experiments/sice1_evaluation/raw_logs/pilot2_audit_chain.jsonl`
  Status: 14 blocks, verified contiguous. Final block hash:
  `3ffcdae08f942f8ee0c20c69850a8a0bf7957a41702fd14d3d7e85499b0bb0c8`
  Intermediate evaluation data digest (sealed into block 14 before serialization):
  `c84e5a134df20dc64dcb57feb538a2048c456c384e92cec9a9511fdbaa0b2f0d`
  Final saved canonical benchmark JSON (`pilot2_summary_results.json`) SHA-256:
  `ad8577f203ff308eeab5eace785a79fc139b97422b43df9245de3036a717bae3`
- **Canonical Benchmark CSV Datasets (`benchmark_data/`):**
  Tabular exports embedded directly into publication PDF and repository:
  - `benchmark_sice_pilot1_batteries.csv` (11 component batteries)
  - `benchmark_sice_pilot1_battery2_paragraphs.csv` (10 Kuhn bipartite paragraphs)
  - `benchmark_sice_pilot2_per_world_lifecycle.csv` (10 procedural worlds lifecycle)
  - `benchmark_sice_pilot2_aggregate_metrics.csv` (Macro scorecard metrics)

7 Hardware Specifications & Benchmarking Methodology

To maximize scientific transparency, reliability, and reproducibility, the host environment, concurrency configuration, power modeling, and latency measurement methodologies are formally specified:

Hardware Dimension	Host Specification	Benchmarking Constraint & Methodology
Processor (CPU)	AMD Ryzen 9 9950X (Zen 5)	16 physical cores, 32 threads, 4.3 GHz base / up to 5.7 GHz boost.
Host Memory (RAM)	64 GB DDR5 (6,000 MT/s)	Maximum sampled process RSS: **443.18 MB** (sampled maximum, not continuous high-water mark).
Operating System	Windows 11 Pro 64-bit (26200)	Execution environment: CPython 3.13.13, clean virtualenv.
Core Concurrency	Single-Threaded OpenMP	Executed with `OMP_NUM_THREADS=1`. Process-level CPU affinity was not independently established.
Release 2.0 Contrast	16-Core / 32-Thread Parallel Cohort	Multiprocessing cohort benchmark scaled across all 16 physical cores and 32 threads (6.14s).
Power Attribution	Not Directly Measured	Electrical power consumption was not directly measured; executed locally on consumer CPU without discrete GPU.
Measurement Tooling	High-Resolution Telemetry	`time.perf_counter()`, OS process sampling, 0.041 ms interoceptive tick.

Table 6: Host test system specifications and execution constraints.

7.1 Internal Realization-Path Latency Measurement Methodology

The reported realization latency characterizes internal algorithmic path latency rather than full user-query-to-response round-trip time. It was empirically profiled using `time.perf_counter()` across 1,000 synthetic queries within `test_multi_turn_dialogue.py` on the host CPU after warm-up:

- **Syntactic Parsing & Morpheme Resolution:** 0.18 ms (subword attractor settling in $\mathbb{R}^{64}$).
- **Centering Theory Discourse Resonance:** 0.42 ms (forward/backward-looking center ranking C_f, C_b, Grosz et al. 1995).
- **CSCG Viterbi Factor-Graph Path Inference:** 1.45 ms (max-product clone transition search, Viterbi 1967).
- **Surface String Realization:** 0.28 ms (syntactic token emission).
- **Component Sum & Percentiles:** Sum of component stages: 2.33 ms. End-to-end realization-path percentiles across individual queries: median ($p50$) = **2.41** ms, $p95$ = **6.80** ms, $p99$ = **11.90** ms (within the 15 ms internal budget). Process startup, OS I/O, IPC, and network serialization were excluded.

8 Component Maturity Classification

To provide an unambiguous picture of the architecture's maturity, all components are classified into three operational tiers:

Maturity Tier	Architectural Modules	Operational Status & Empirical Defense
Tier 1: Operational	• Sparse CSR Cortical Tiling • Kuhn Bipartite Text Matcher • Buckingham Π SVD Filter • EarthosGuard Sovereign Kernel • Cryptographic Audit Chain	Operational components, zero external ML dependencies. Bit-level determinism verified under identical build/seed on the host platform. Sparse CSR hash lookup ($p50 = 0.40\,\mu$s) sampled across 1,000 dictionary/array operations.
Tier 2: Integrated Prototypes	• Paired Procedural Worlds Bank • Diagnostic SCM Clamps ($B = 30$) • Gentner Topological Analogy • Blunder Attractor Quenching • SHY Computational Sleep • Prefrontal Working Memory ($N = 8$)	Integrated prototypes internally validated under preregistered SICE-1 protocols across 10 worlds; demonstrates integrated multi-stage developmental lifecycle without harness resets.
Tier 3: Research-Stage	• Out-of-Process IPC Sandbox • Continuous Matched Controls • Full-DAG Continuous Scoring • DIVA Motor Predictive Coding • Hippocampal Episodic Dialogue	Research-stage components scheduled for Pilot-3 (Release 3.1) and Release 4.0; addresses process-level sandboxing, continuous control history alignment, and token-by-token vocal tract predictive coding (Tourville & Guenther 2011).

Table 7: Three-tier component maturity classification.

9 Conclusion: The Sovereign Live Mind Era

With **Public Release 3.0**, Synapse establishes:

1. A rigorous component engineering baseline (Pilot-1) proving that non-neural biological mechanisms perform causal induction, physical invariant discovery, and text distillation.
2. An integrated cognitive organism benchmark (Pilot-2) proving that perception, active experimentation, structural analogy, interactive planning, and sleep consolidation operate as a unified system in continuous execution.
3. Local CPU execution without external LLM inference APIs, transformer inference services, or discrete GPU acceleration.

Both Pilot-1 and Pilot-2 are internally validated against the preregistered evaluation battery, cryptographically hash-chained, and publicly released under the open SICE-1 evaluation specification.

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